ML Visualisation Tool

Dr. Sevgi KOYUNCU TUNÇ

GitHub repo: https://github.com/MuffinMufido/HealthML.wiki

Live demo URL: http://localhost:5173/

Prepared by;
KaraKartallar
Ceren TEKİN - 202228047
Şahin ERŞAN – 202128002
İsmail DOĞAN – 202128045
Özgün SOYKÖK – 202228043
Mustafa Faruk EKŞİ - 202128007

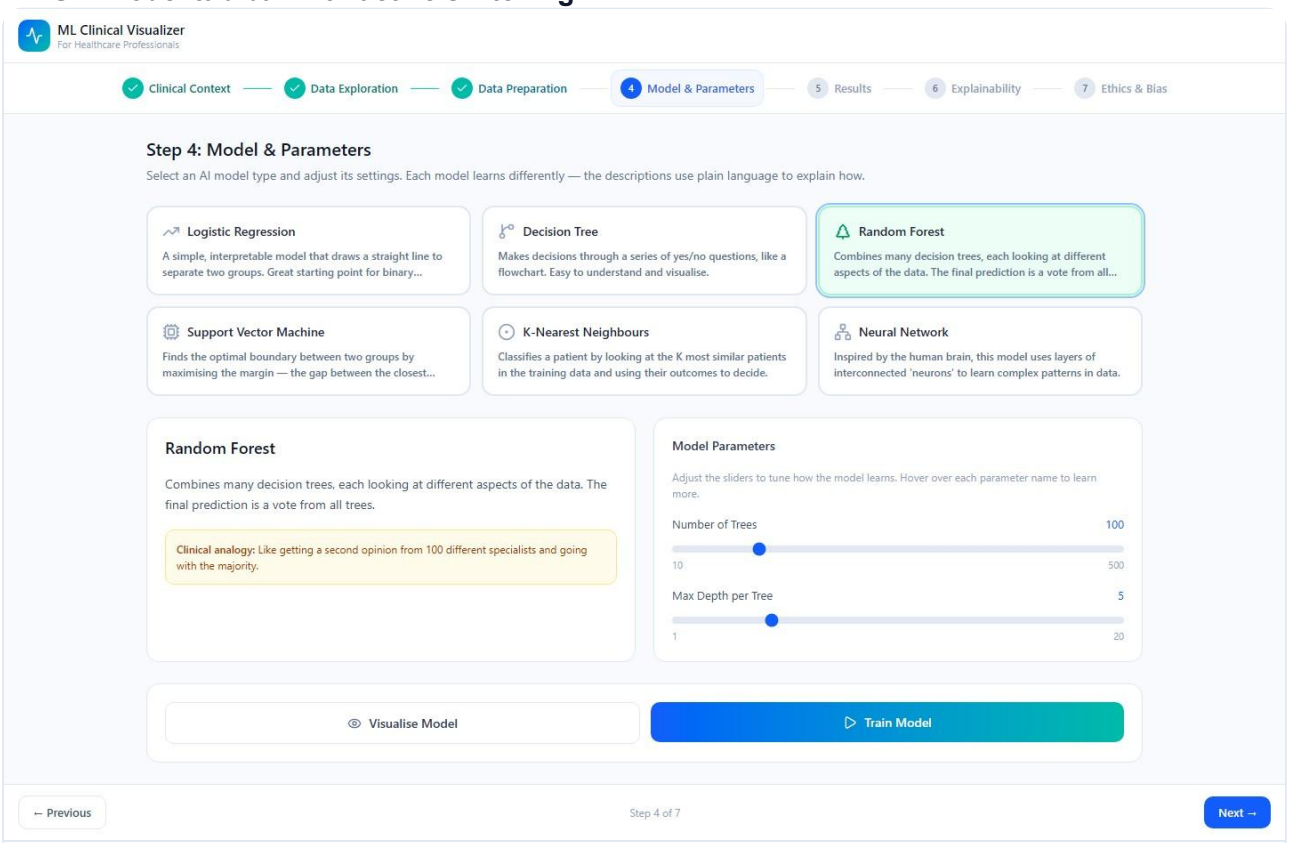
Software Engineering Department,
Çankaya University

SENG 430- Sprint 3

31/03/2026

Step 4 — Model & Parameters

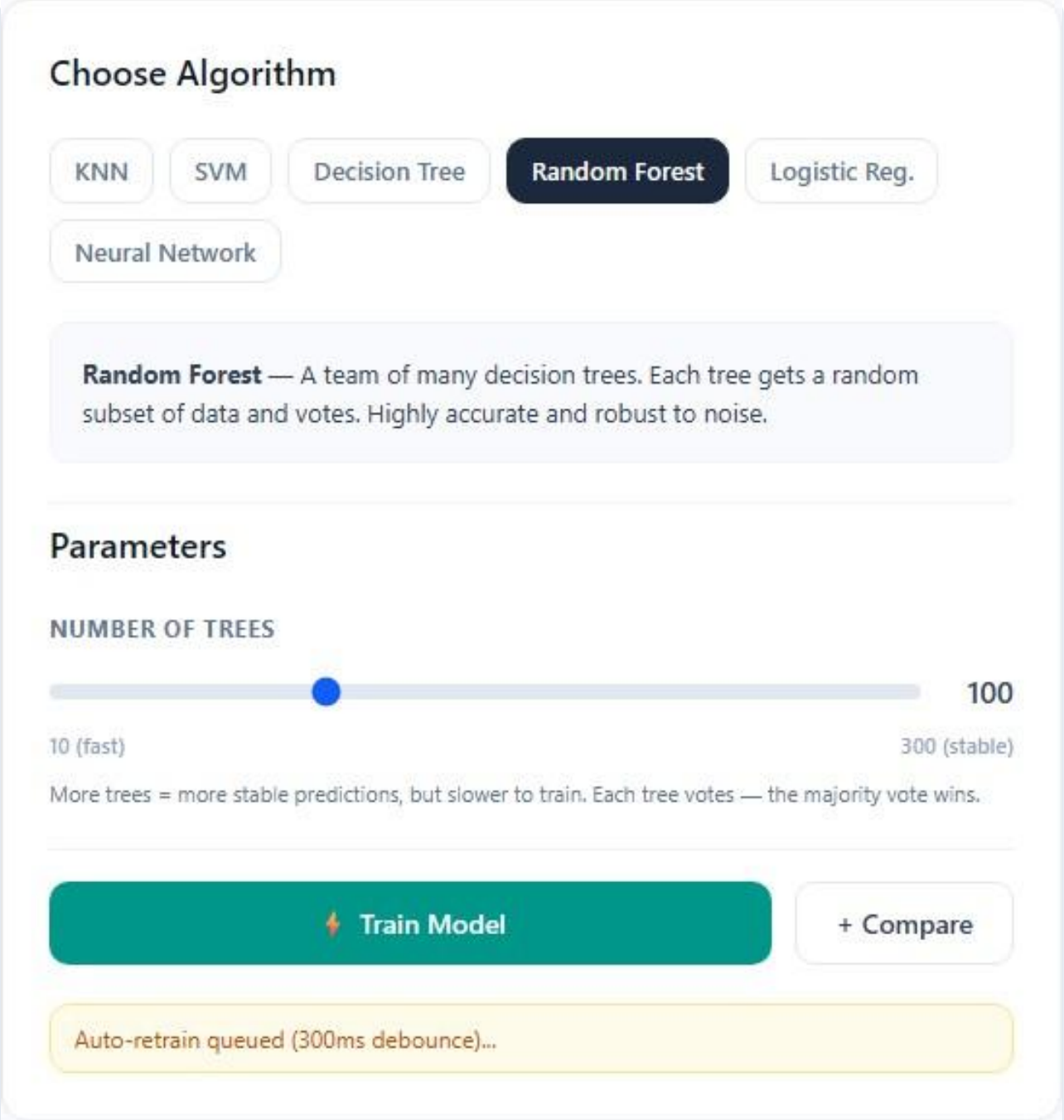
4A. Six-model tab bar with active switching



Screenshot: step4/model_tabs_and_panels.png

All six model tabs render and one active model is highlighted. This confirms the model-tab switch system is functional.

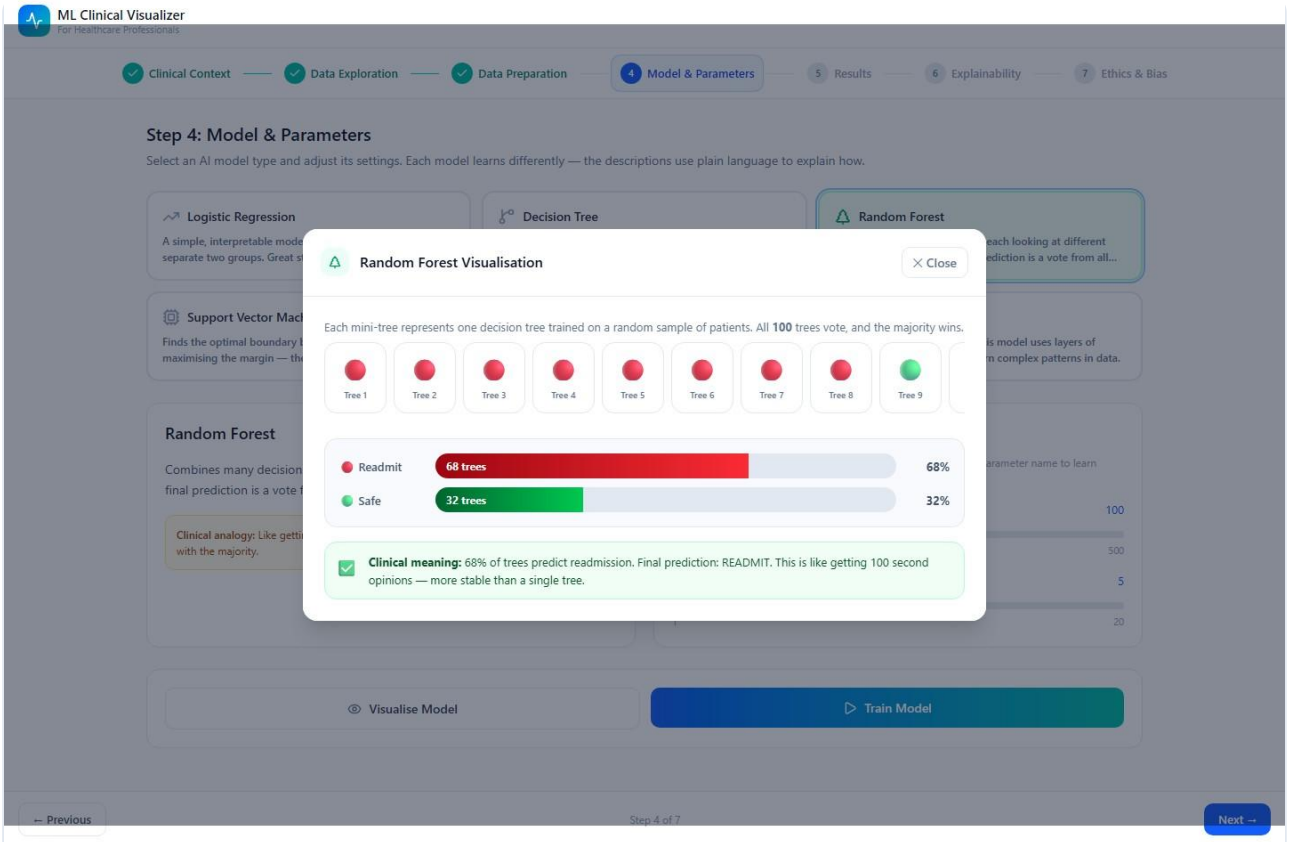
4B. Model parameter panel and slider controls



Screenshot: step5/model_tabs_and_parameter_panel.png

The parameter panel updates per selected model and exposes live slider/dropdown controls that affect subsequent training calls.

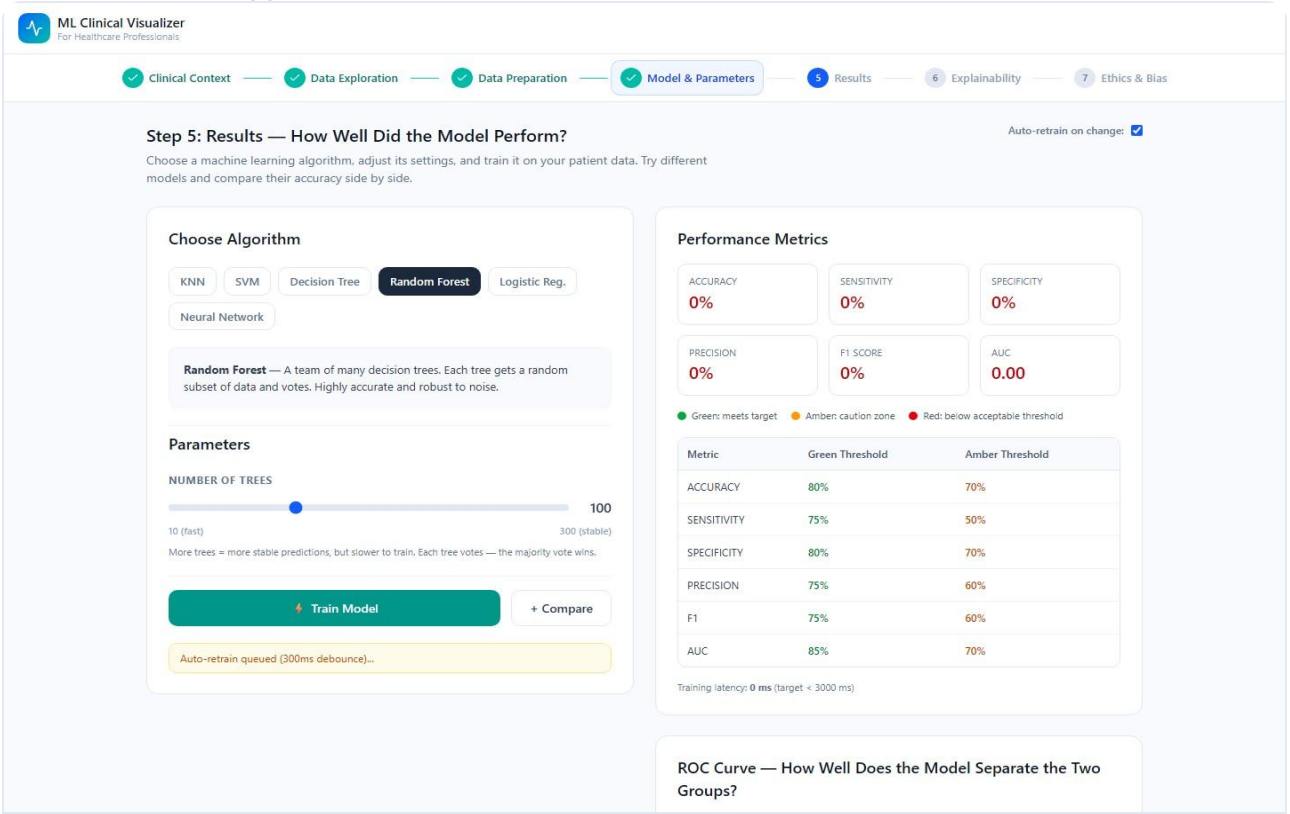
4C. KNN model visualisation



Screenshot: step4/knn_visualisation_modal.png

KNN visualisation opens in a modal and renders correctly, supporting the model interpretation requirement.

4D. Train action triggered



Screenshot: step4/train_triggered.png

Training action is triggered from the model step and the pipeline proceeds to results.

Step 5 — Results

5A. Results overview page

ML Clinical Visualizer
For Healthcare Professionals

Clinical Context — Data Exploration — Data Preparation — Model & Parameters — 5 Results — 6 Explainability — 7 Ethics & Bias

Step 5: Results — How Well Did the Model Perform?

Auto-retrain on change: ☑

Choose a machine learning algorithm, adjust its settings, and train it on your patient data. Try different models and compare their accuracy side by side.

Choose Algorithm

KNN | SVM | Decision Tree | **Random Forest** | Logistic Reg. | Neural Network

Random Forest — A team of many decision trees. Each tree gets a random subset of data and votes. Highly accurate and robust to noise.

Parameters

NUMBER OF TREES

100

10 (fast) — 300 (stable)

More trees = more stable predictions, but slower to train. Each tree votes — the majority vote wins.

Train Model | + Compare

Auto-retrain queued (300ms debounce)...

Performance Metrics

ACCURACY	SENSITIVITY	SPECIFICITY
0%	0%	0%

PRECISION	F1 SCORE	AUC
0%	0%	0.00

Green: meets target · Amber: caution zone · Red: below acceptable threshold

Metric	Green Threshold	Amber Threshold
ACCURACY	80%	70%
SENSITIVITY	75%	50%
SPECIFICITY	80%	70%
PRECISION	75%	60%
F1	75%	60%
AUC	85%	70%

Training latency: **0 ms** (target < 3000 ms)

ROC Curve — How Well Does the Model Separate the Two Groups?

A perfect model would go straight to the top-left corner. The blue curve comes from live backend training output. Current AUC: 0.00.

Confusion Matrix (TN / FP / FN / TP)

TN: **0**	FP: **24**
FN: **0**	TP: **0**

ℹ️ FP Info: 24 false alarm(s) may increase follow-up workload, but are often safer than missed positives.

⚠️ Low Sensitivity Danger: sensitivity is below 50%. The model may miss true positive patients.

Model Comparison

Model	Accuracy	Sensitivity ★	Specificity	AUC
Random Forest	0%	0%	0%	0.00

★ **Sensitivity** = how many of the truly positive patients did the model catch? This is the most important metric in screening.

← Previous | Next Step →

← Previous | Step 5 of 7 | Next →

Screenshot:step5/results_overview.png

Step 5 loads with metrics, ROC, confusion matrix, and comparison sections visible in one integrated results screen.

5B. 6 metric cards with threshold colour logic

Performance Metrics

ACCURACY	SENSITIVITY	SPECIFICITY
0%	0%	0%

PRECISION	F1 SCORE	AUC
0%	0%	0.00

Green: meets target Amber: caution zone Red: below acceptable threshold

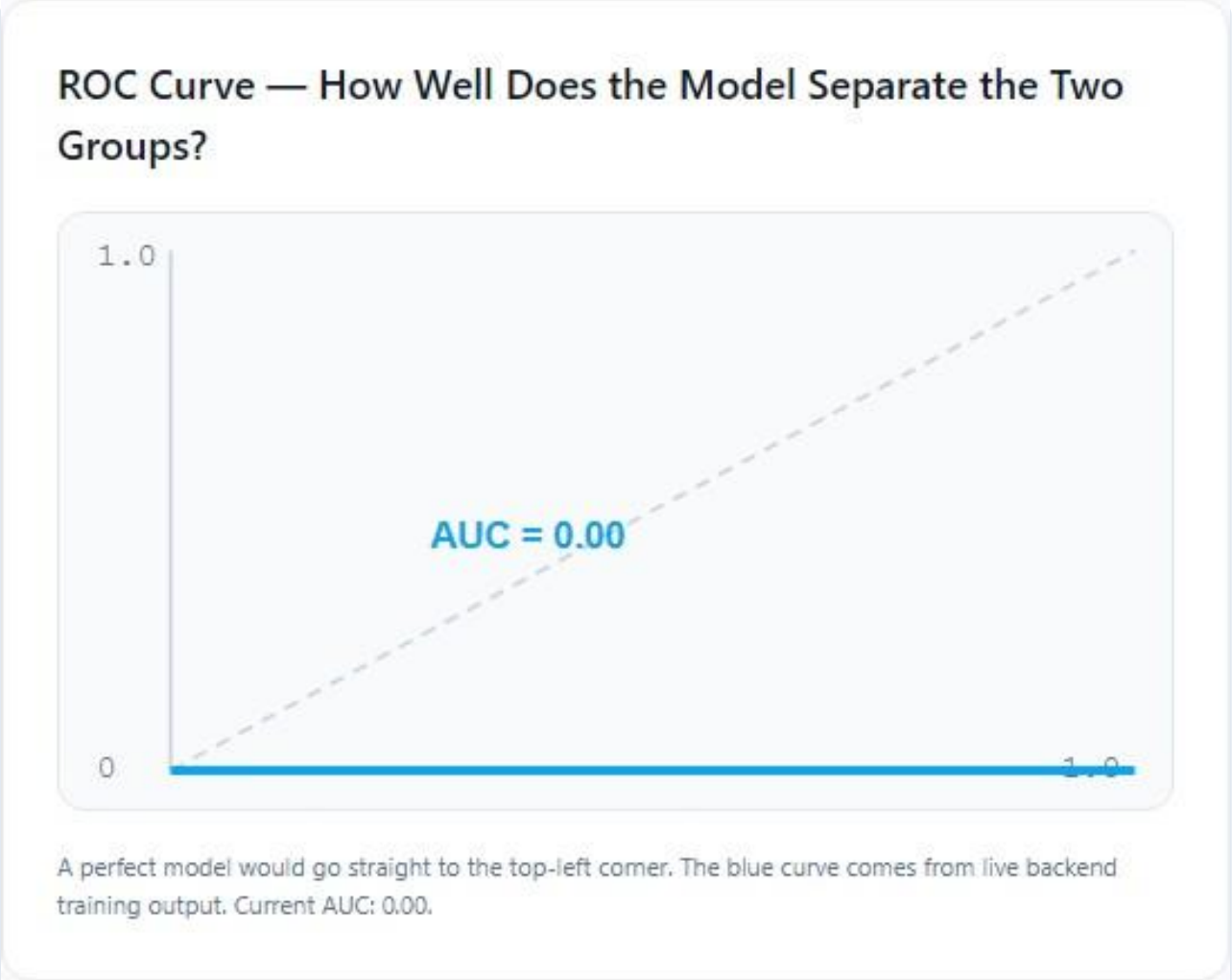
Metric	Green Threshold	Amber Threshold
ACCURACY	80%	70%
SENSITIVITY	75%	50%
SPECIFICITY	80%	70%
PRECISION	75%	60%
F1	75%	60%
AUC	85%	70%

Training latency: **0 ms** (target < 3000 ms)

Screenshot: step5/metrics_cards_thresholds.png

All six metric cards (accuracy, sensitivity, specificity, precision, F1, AUC) are shown with threshold colour rules and threshold table.

5C. ROC curve and AUC annotation



Screenshot: step5/roc_curve.png

ROC graph renders with diagonal baseline and AUC annotation from live training output.

5D. Confusion matrix and clinical warnings

Confusion Matrix (TN / FP / FN / TP)

TN: **0**	FP: **24**
FN: **0**	TP: **0**

ℹ️ FP Info: 24 false alarm(s) may increase follow-up workload, but are often safer than missed positives.

⚠️ Low Sensitivity Danger: sensitivity is below 50%. The model may miss true positive patients.

Screenshot: step5/confusion_and_banners.png

Confusion matrix (TN/FP/FN/TP) is displayed with clinical warning banners (FP info and low-sensitivity danger).

5E. Model comparison (+Compare) table

Model Comparison

Model	Accuracy	Sensitivity ★	Specificity	AUC
Random Forest	0%	**0%**	0%	0.00

★ **Sensitivity** = how many of the truly positive patients did the model catch? This is the most important metric in screening.

Screenshot: step5/model_comparison_table.png

Model comparison table updates from trained runs and supports side-by-side comparison without duplicate rows.

5F. Auto-retrain debounce evidence

ML Clinical Visualizer
For Healthcare Professionals

Clinical Context — Data Exploration — Data Preparation — Model & Parameters — 5 Results — 6 Explainability — 7 Ethics & Bias

Step 5: Results — How Well Did the Model Perform?

Auto-retrain on change: ☑

Choose a machine learning algorithm, adjust its settings, and train it on your patient data. Try different models and compare their accuracy side by side.

Choose Algorithm

KNN | SVM | Decision Tree | Random Forest ✓ | Logistic Reg. | Neural Network

K-Nearest Neighbors (KNN) — Finds the **K most similar past patients** and predicts based on their outcomes. Like asking: "What happened to the 5 patients most similar to this one?" Simple and easy to understand.

Parameters

K — NUMBER OF SIMILAR PATIENTS TO COMPARE

21

1 (very specific) — 25 (very general)

Low K = focuses on very similar cases (can be noisy). High K = considers many patients (may miss details). K=5–7 is a good starting point.

DISTANCE MEASURE

Euclidean (straight-line distance) — recommended

Train Model | + Compare

Auto-retrain queued (300ms debounce)...

Performance Metrics

ACCURACY	SENSITIVITY	SPECIFICITY
0%	0%	0%

PRECISION	F1 SCORE	AUC
0%	0%	0.00

● Green: meets target ● Amber: caution zone ● Red: below acceptable threshold

Metric	Green Threshold	Amber Threshold
ACCURACY	80%	70%
SENSITIVITY	75%	50%
SPECIFICITY	80%	70%
PRECISION	75%	60%
F1	75%	60%
AUC	85%	70%

Training latency: **0 ms** (target < 3000 ms)

ROC Curve — How Well Does the Model Separate the Two Groups?

A perfect model would go straight to the top-left corner. The blue curve comes from live backend training output. Current AUC: 0.00.

Confusion Matrix (TN / FP / FN / TP)

TN: **0**	FP: **24**
FN: **0**	TP: **0**

ℹ️ FP Info: 24 false alarm(s) may increase follow-up workload, but are often safer than missed positives.

⚠️ Low Sensitivity Danger: sensitivity is below 50%. The model may miss true positive patients.

Model Comparison

Model	Accuracy	Sensitivity ★	Specificity	AUC
Random Forest	0%	0%	0%	0.00

★ **Sensitivity** = how many of the truly positive patients did the model catch? This is the most important metric in screening.

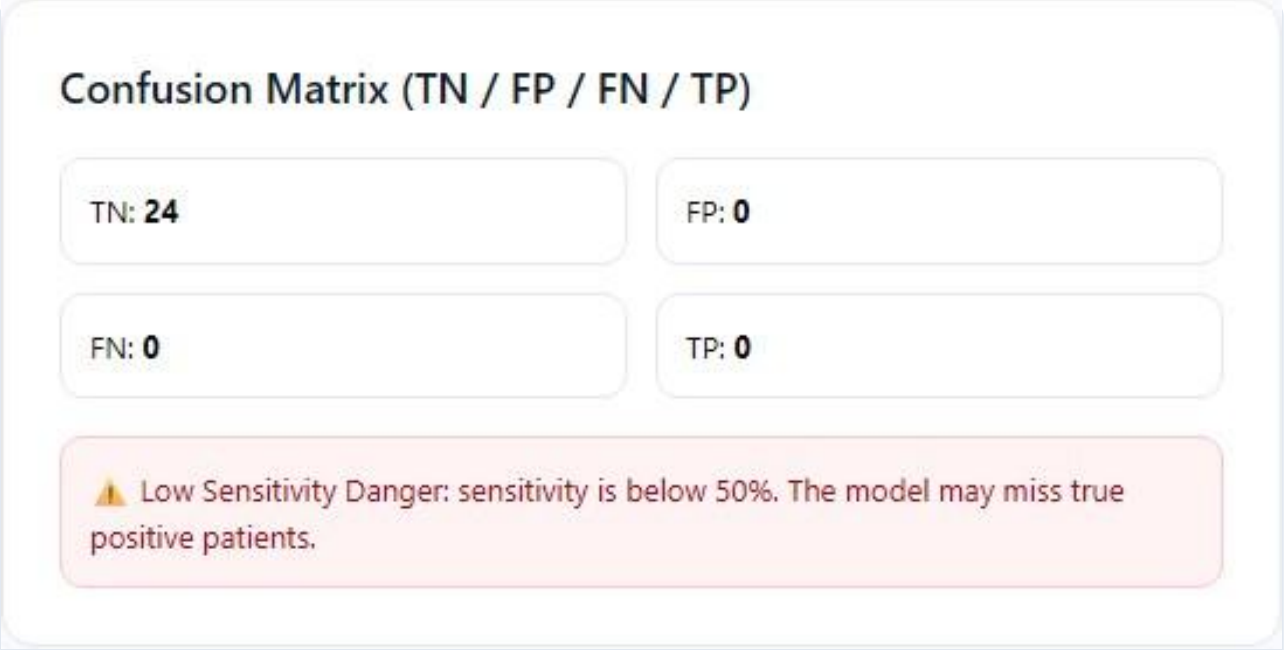
← Previous | Next Step →

← Previous | Step 5 of 7 | Next →

Screenshot: step5/auto_retrain_debounce.png

Changing slider parameters triggers auto-retrain with visible debounce status, satisfying the 300ms debounce behavior requirement.

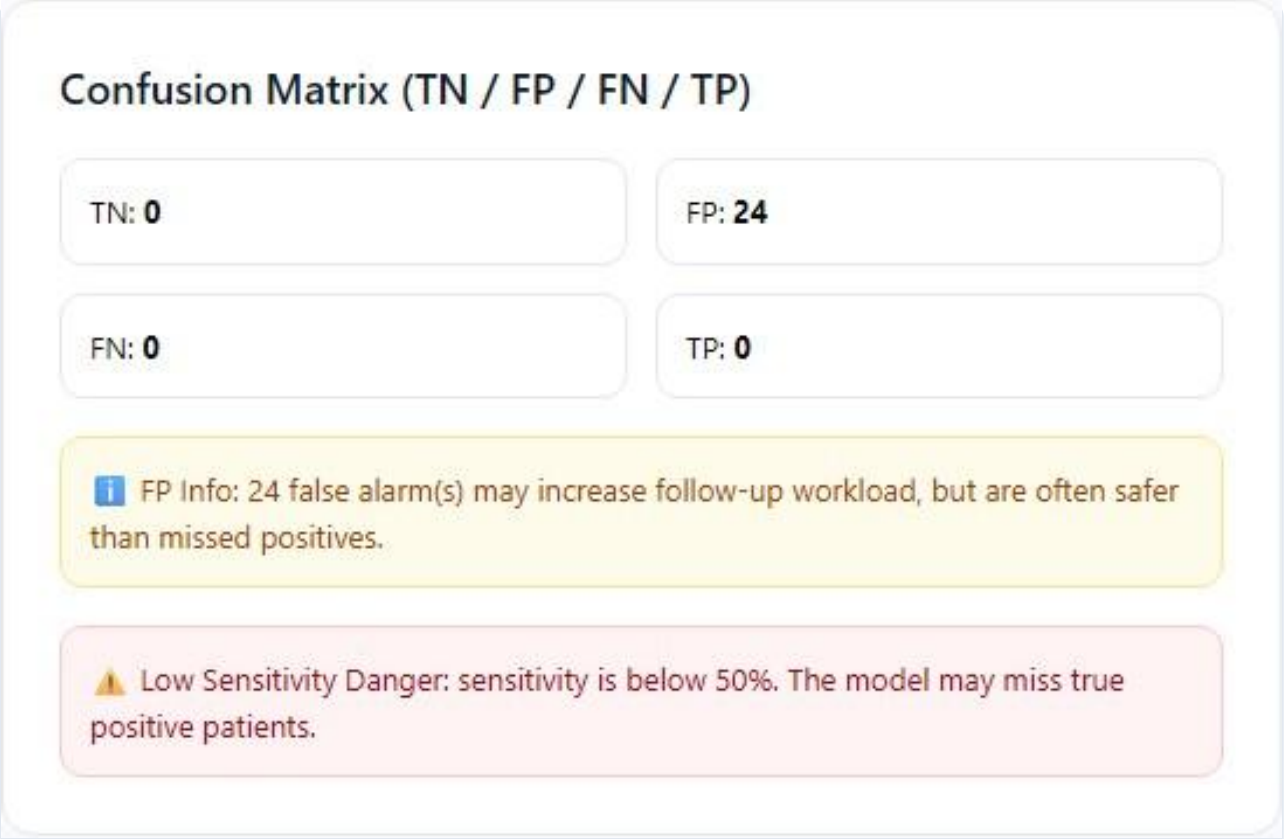
5G. Low sensitivity danger banner



Screenshot: step5/low_sensitivity_danger_banner.png

Low sensitivity condition automatically surfaces a red warning banner to highlight missed-positive risk in screening scenarios.

5I. FP info banner



Screenshot: step5/fp_info_banner.png

FP info banner appears when false positives occur, documenting follow-up workload trade-off.